MDPI

Article

Supplementary Material: WRF Cold-Pool Sensitivities Relevant to Winter-Ozone Modelling in the Uinta Basin

John R. Lawson [1,2,*], Michael J. Davies [1], Loknath Dhar [1,3] and Seth N. Lyman [1,3]

[1] Bingham Research Center, Utah State University, Vernal, UT 84078, USA;
[2] Department of Mathematics and Statistics, Utah State University, Logan, UT 84322, USA;
[3] Department of Chemistry and Biochemistry, Utah State University, Logan, UT 84322, USA
* Correspondence: john.lawson@usu.edu

1. Within-X8 d04 Heat-Deficit Transport

X8 is a single multiply confounded context run with 111.111 m nominal d04 grid spacing, 50 vertical levels, and 10 min output. The panels in Figure 1 describe resolved model processes within X8 and do not provide a comparison, independent validation, resolution response, numerical-viability test, or basin-export estimate.

Received:
Revised:
Accepted:
Published:

Copyright: © 2026 by the authors. Submitted to *Air* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license.

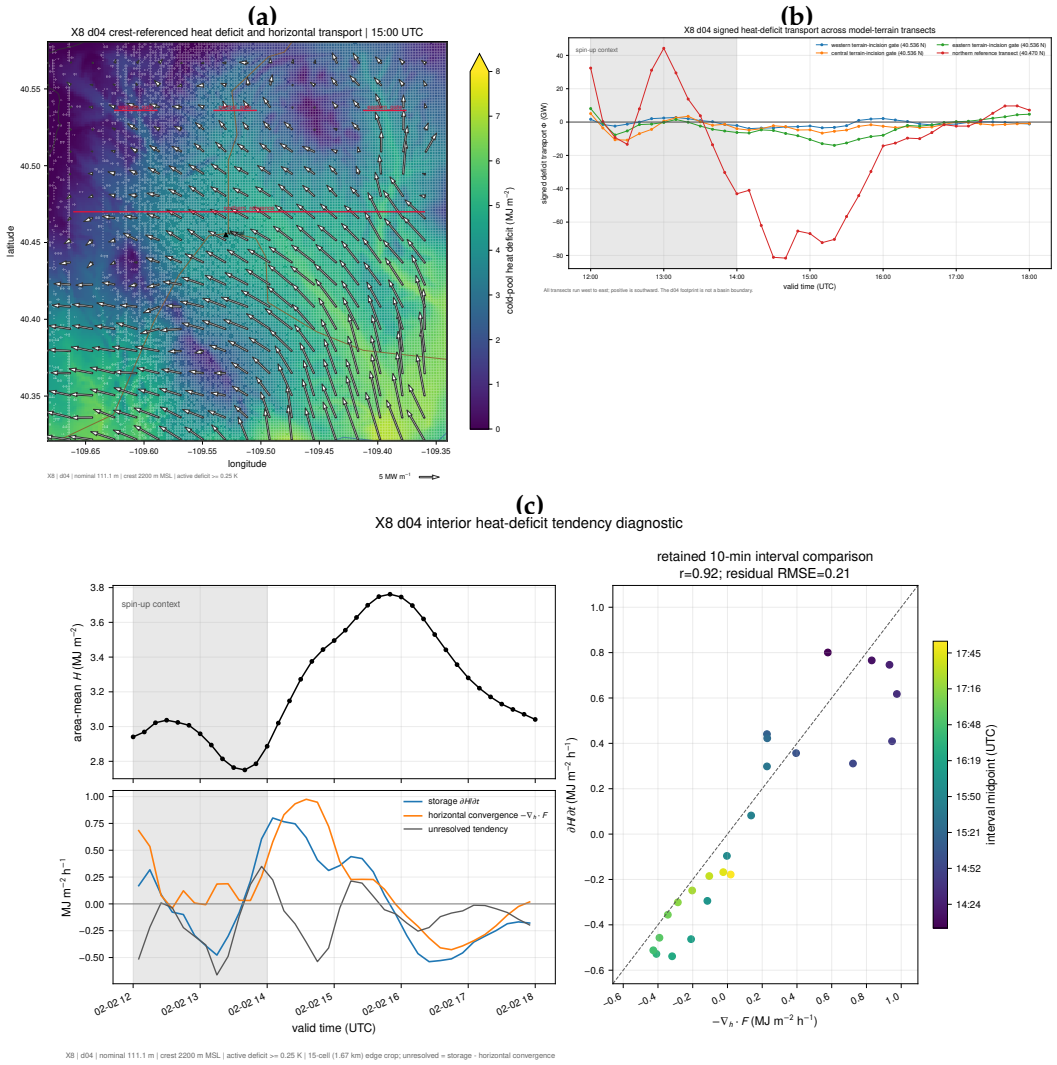


Figure 1. Within-X8 d04 heat-deficit transport and tendency diagnostics. (a) Crest-referenced heat deficit H and horizontal deficit transport **F** at 1500 UTC use the local potential temperature at 2200 m mean sea level as the column reference. The three short lines cross model-terrain incisions, while the longer line is a northern reference transect rather than a basin boundary. (b) Signed transport at 10 min cadence is positive southward and negative northward because every transect runs west to east. The northern reference line reaches -81.61 GW at 1440 UTC before relaxing to $+9.69$ GW at 1750 UTC. (c) Ten-minute finite-difference storage is compared with endpoint-mean horizontal convergence after excluding 15 cells from each d04 edge; their difference is retained as unresolved rather than attributed to a single process. Across retained intervals, storage and horizontal convergence have $r = 0.92$, while the unresolved-tendency root-mean-square error is $0.21\,\mathrm{MJ\,m^{-2}\,h^{-1}}$. The shaded 1200–1400 UTC interval is spin-up context, and none of the panels establishes closed budget accounting, observed canyon drainage, or basin export.

2. Slope-Aware-Radiation Diagnostic

The controlled slope-aware-radiation treatment produced negligible response at the sampled basin-floor columns but an aspect-organised skin-temperature response on steep terrain. The archived shortwave field is a flat-plane quantity and did not express the treatment, while the archive lacks the surface heat-flux terms needed to test the mechanism directly.

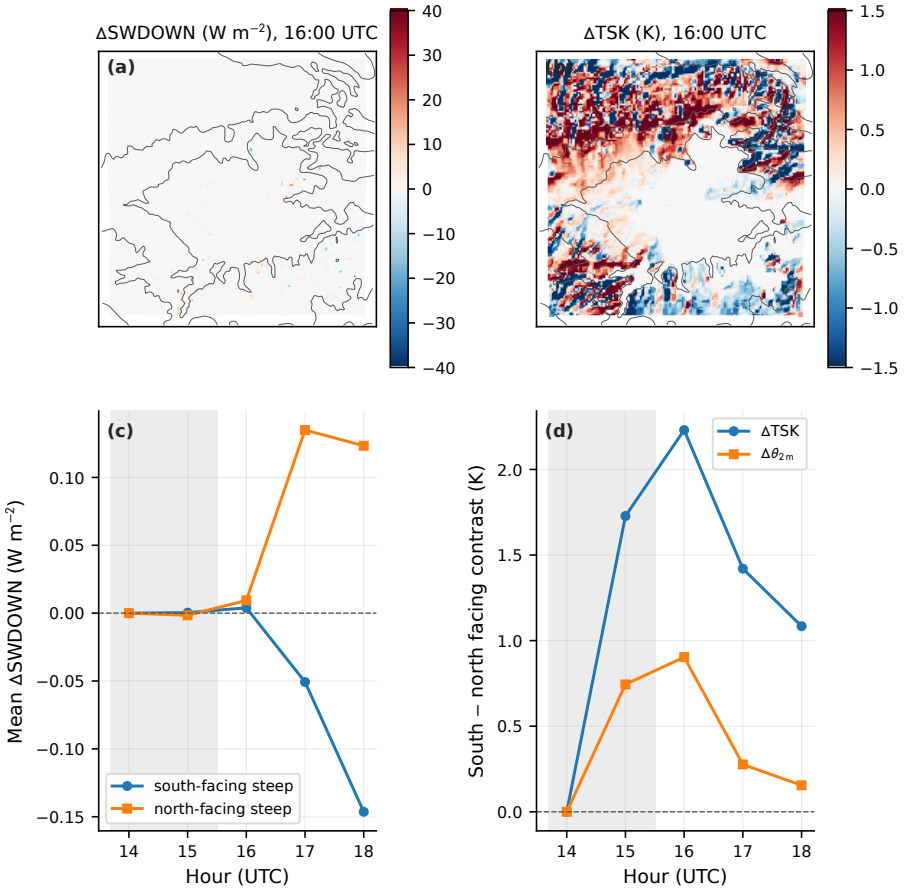


Figure 2. Diagnostic response to slope-aware radiation and terrain shading on the controlled fine-terrain branch. The predeclared screen required a response in the archived flat-plane shortwave field and therefore failed, while skin temperature showed an aspect-organised response on steep terrain. The figure demonstrates that the common station metric can miss a localised model response, but it does not show that slope-aware radiation improved or degraded the cold pool and supports no configuration ranking.